

(a) Problem setup



First-visit visual evidence
road-sign crop written once

(b) Memory write/store

crop path

feature vector

metadata

External memory

inference-time only

(c) Candidate generation

Frozen MatrixGame-2



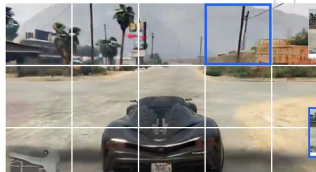
seed1



seed8

candidate seed1 ... seed8
same pool before reranking

(d) Memory read by patch matching



Regular grid patches

Cosine similarity

Feature encoder

best local patch

(e) Candidate reranking



No memory:
seed1

With memory:
seed8

seed1 -> seed8

selected output changes

No object segmentation. Patches are regular grid cells.

ResNet / DINO / CLIP are feature extractors.